REPORT

Forest Fire Detection & Spread Prediction System

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Milestone – 4

Analytics Output & Dashboard Visualization

Overview

Milestone 4 focuses on storing, structuring, and visually presenting the prediction outputs generated by the ML pipeline in Milestone 3. Real-time sensor readings and fire risk predictions are saved in a structured Excel/CSV format and then imported into Microsoft Power BI to build an interactive analytics dashboard. The dashboard visualizes key environmental indicators, fire alert distributions, and geographic location data, enabling operators to monitor forest fire conditions and respond proactively.

Objectives

- Store real-time sensor data and ML prediction outputs in a structured CSV and Excel format.
- Import the structured data into Microsoft Power BI for visualization.
- Develop dashboard visuals including pie charts, donut charts, bar charts, scatter plots, and gauge indicators.
- Display fire alert distribution, temperature trends, humidity relationships, and geographic coordinates.
- Enable non-technical stakeholders to interpret fire risk conditions at a glance.

Section 1: Structured Data Output

After each prediction session, the live prediction script (predict_live.py) exports all cycle results to two files: forest_fire_predictions.csv and forest_fire_predictions.xlsx. These files serve as the data source for the Power BI dashboard.

Output File Columns

Column	Description
Cycle	Prediction cycle number (1 to NUM_CYCLES)
Timestamp	UTC timestamp of the ThingSpeak reading (e.g., 2026-04-09T17:51:23Z)
Final Temp (°C)	Fused temperature: average of DHT11 sensor and OpenWeatherMap API values
Final Hum (%)	Fused humidity: average of DHT11 sensor and OpenWeatherMap API values

Wind	Wind speed and direction from OpenWeatherMap API (km/h + compass direction)
Dryness	Dryness Index = Final Temp minus Final Hum; key fire-risk feature
Gas	MQ-2 gas sensor status: HIGH or LOW
Flame	Flame sensor status: YES or NO
Alert	System fire alert category (NORMAL, FIRE DETECTED, CRITICAL, EXTREME FIRE RISK, FIRE ALERT)
Model	ML model classification output: LOW, MEDIUM, HIGH, or FIRE
Latitude	GPS latitude of the monitored forest location
Longitude	GPS longitude of the monitored forest location
Location	Human-readable location name (e.g., Nainital Forest, Uttarakhand)

Sample Data Records

The screenshot below shows the first 30 rows of the exported Excel file as loaded into the Power BI data source. Each row corresponds to one prediction cycle, with sensor readings, derived values, alert classification, and geographic coordinates.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Cycle	Timestamp	Final Temp	Final Hum	Wind	Dryness	Gas	Flame	Alert	Model	Latitude	Longitude	Location		
1	2026-04-0	43.8	54.88	13.0 S	-11.08	HIGH	NO	NORM	LOW	29.386	79.4819	Nainital Forest, Uttarakhand		
2	2026-04-0	44.46	41.22	28.4 SE	3.24	LOW	YES	FIRE D	FIRE	29.39554	79.49063	Nainital Forest, Uttarakhand		
3	2026-04-0	19.61	39.53	16.2 N	-19.92	LOW	YES	FIRE D	FIRE	29.3815	79.46553	Nainital Forest, Uttarakhand		
4	2026-04-0	39.79	44.54	17.8 NW	-4.75	HIGH	YES	CRITIC	FIRE	29.38719	79.48303	Nainital Forest, Uttarakhand		
5	2026-04-0	16.02	24.12	39.6 SE	-8.1	HIGH	NO	NORM	MEDIUM	29.35408	79.43545	Nainital Forest, Uttarakhand		
6	2026-04-0	24.83	63.9	28.5 E	-39.07	HIGH	NO	NORM	LOW	29.44036	79.41012	Nainital Forest, Uttarakhand		
7	2026-04-0	28.22	67.61	3.4 SW	-39.39	LOW	YES	FIRE D	FIRE	29.40036	79.45944	Nainital Forest, Uttarakhand		
8	2026-04-0	35.79	18.49	22.5 SW	17.3	HIGH	NO	NORM	LOW	29.41075	79.43673	Nainital Forest, Uttarakhand		
9	2026-04-0	30.76	75.59	22.3 W	-44.83	LOW	NO	NORM	LOW	29.42401	79.47599	Nainital Forest, Uttarakhand		
10	2026-04-0	32.43	68.02	8.1 NW	-35.59	LOW	YES	FIRE D	FIRE	29.43521	79.48193	Nainital Forest, Uttarakhand		
11	2026-04-0	33.58	22.89	3.2 W	10.69	LOW	NO	NORM	LOW	29.39601	79.42099	Nainital Forest, Uttarakhand		
12	2026-04-0	20.81	52.48	37.6 W	-31.67	HIGH	YES	CRITIC	FIRE	29.37648	79.46021	Nainital Forest, Uttarakhand		
13	2026-04-0	42.39	70.4	36.4 SW	-28.01	HIGH	NO	NORM	MEDIUM	29.36981	79.47985	Nainital Forest, Uttarakhand		
14	2026-04-0	27.48	31.26	17.7 NW	-3.78	HIGH	YES	CRITIC	FIRE	29.42385	79.45459	Nainital Forest, Uttarakhand		
15	2026-04-0	25.44	71.34	2.9 SW	-45.9	LOW	NO	NORM	MEDIUM	29.44033	79.46909	Nainital Forest, Uttarakhand		
16	2026-04-0	40.38	85.26	32.7 W	-44.88	HIGH	YES	CRITIC	FIRE	29.41716	79.46547	Nainital Forest, Uttarakhand		
17	2026-04-0	33.84	37.87	37.0 NW	-4.03	LOW	YES	FIRE D	FIRE	29.39656	79.43991	Nainital Forest, Uttarakhand		
18	2026-04-0	31.94	61.38	21.7 W	-29.44	HIGH	NO	NORM	MEDIUM	29.42249	79.42541	Nainital Forest, Uttarakhand		
19	2026-04-0	19.96	45.55	37.7 SE	-25.59	HIGH	NO	NORM	LOW	29.36513	79.46866	Nainital Forest, Uttarakhand		
20	2026-04-0	36.26	45.25	39.3 E	-8.99	LOW	NO	NORM	MEDIUM	29.36719	79.49058	Nainital Forest, Uttarakhand		
21	2026-04-0	30.92	66.62	35.1 S	-35.7	LOW	YES	FIRE D	FIRE	29.37866	79.50388	Nainital Forest, Uttarakhand		
22	2026-04-0	33.39	62.03	16.8 S	-28.64	HIGH	NO	NORM	MEDIUM	29.41251	79.4685	Nainital Forest, Uttarakhand		
23	2026-04-0	39.76	80.03	23.4 NW	-40.27	HIGH	YES	CRITIC	FIRE	29.37045	79.41044	Nainital Forest, Uttarakhand		
24	2026-04-0	29.27	75.48	39.7 NE	-46.21	LOW	YES	FIRE D	FIRE	29.39485	79.41501	Nainital Forest, Uttarakhand		
25	2026-04-0	44.21	28.72	31.6 NE	15.49	HIGH	NO	NORM	MEDIUM	29.37849	79.44998	Nainital Forest, Uttarakhand		
26	2026-04-0	39.77	88.99	30.1 W	-49.22	LOW	NO	NORM	MEDIUM	29.37228	79.40729	Nainital Forest, Uttarakhand		
27	2026-04-0	43.78	59.75	31.7 N	-15.97	LOW	YES	FIRE D	FIRE	29.43915	79.44302	Nainital Forest, Uttarakhand		
28	2026-04-0	29.46	57.05	22.2 NW	-27.59	HIGH	YES	CRITIC	FIRE	29.35972	79.45858	Nainital Forest, Uttarakhand		
29	2026-04-0	27.76	65.48	33.8 NE	-37.72	HIGH	YES	CRITIC	FIRE	29.4211	79.49462	Nainital Forest, Uttarakhand		
30	2026-04-0	20.69	40.7	21.1 N	-20.01	LOW	YES	FIRE D	FIRE	29.42611	79.49932	Nainital Forest, Uttarakhand		

Section 2: Alert Classification Logic

The Alert column in the output file is derived from a combination of sensor readings and the ML model output. The following table describes the five alert categories used in the dashboard:

Alert Category	Trigger Condition
NORMAL	No gas, no flame, and ML model predicts LOW or MEDIUM risk
FIRE ALERT (SMOKE + HIGH)	Gas sensor HIGH and ML model predicts HIGH risk — elevated danger
FIRE DETECTED (FLAME)	Flame sensor active (YES) — direct fire presence confirmed
CRITICAL: VERIFIED FIRE	Both gas HIGH and flame YES — flame and smoke both confirmed simultaneously
EXTREME FIRE RISK (ML)	ML model predicts FIRE class — based on extreme dryness, temperature, and wind conditions

Section 3: Power BI Dashboard

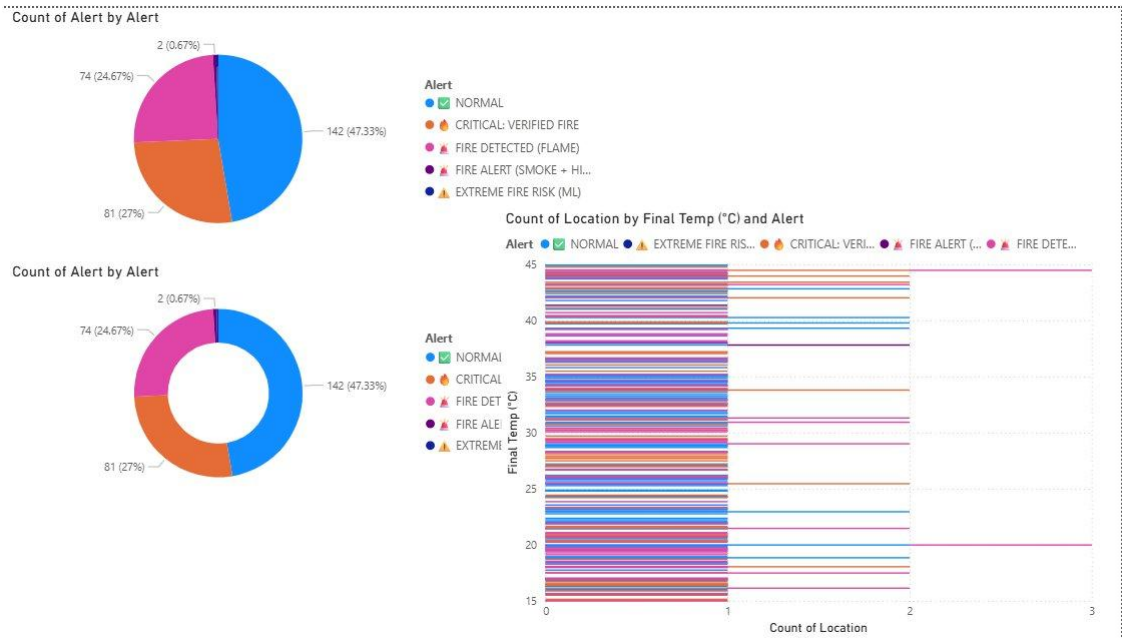
The Power BI dashboard was developed by importing the Excel prediction output file directly as a data source. The dashboard contains five visual components spread across two report pages: alert distribution charts, a temperature gauge, a temperature-timestamp bar chart, a temperature-humidity scatter plot, and summary KPI cards.

Dashboard Visual 1: Alert Distribution — Pie Chart and Donut Chart

Two complementary charts display the proportion of each alert category across all prediction cycles. The pie chart shows filled segments while the donut chart highlights the relative share using a ring format. Both charts display the same data and use the Alert column as the legend field.

Observed alert distribution across all 300 prediction records:

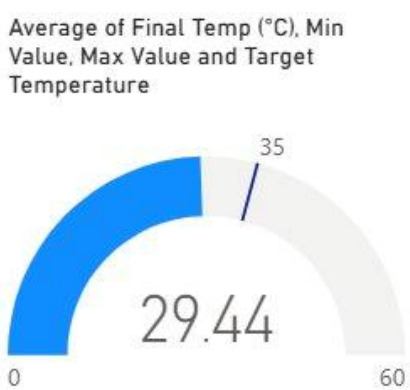
Alert Category	Count	Percentage
NORMAL	142	47.33%
CRITICAL: VERIFIED FIRE	81	27.00%
FIRE DETECTED (FLAME)	74	24.67%
EXTREME FIRE RISK (ML)	2	0.67%
FIRE ALERT (SMOKE + HIGH)	1	0.33%



Dashboard Visual 2: Temperature Gauge

A gauge visual displays the average Final Temperature across all prediction cycles. The gauge is configured with a minimum of 0°C, a target line at 35°C (representing the threshold above which conditions are considered elevated fire risk), and a maximum of 60°C.

The current average Final Temperature reading of 29.44°C is displayed inside the gauge arc. This reading falls below the 35°C target line, indicating that the overall dataset skews toward lower-risk temperature conditions, consistent with the 47.33% NORMAL alert proportion.



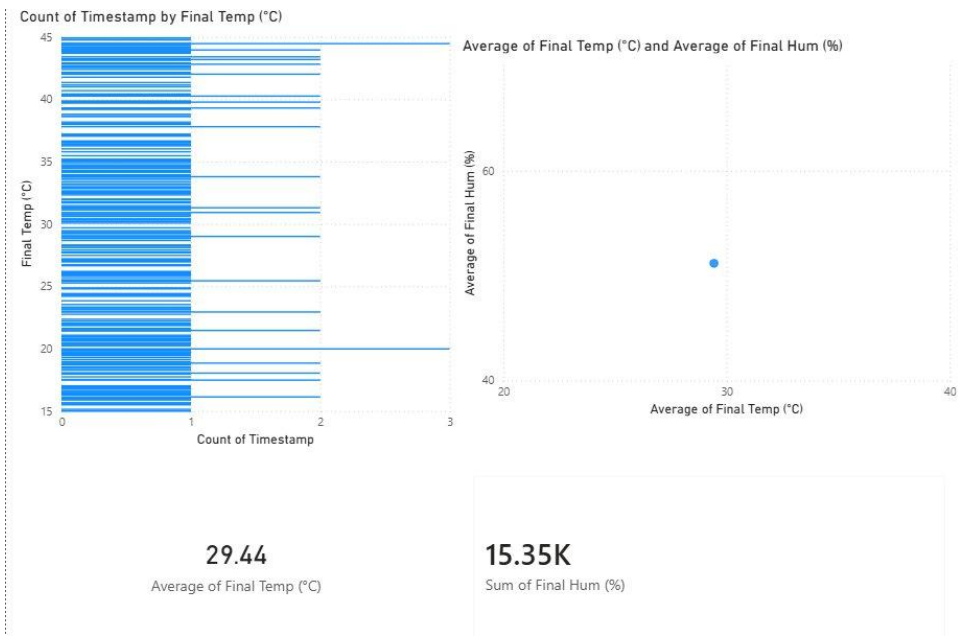
Dashboard Visual 3: Temperature Distribution Bar Chart and Scatter Plot

Two additional visuals are placed side by side on the second report page. The horizontal bar chart on the left shows the count of timestamps for each distinct Final Temperature value, revealing how frequently each temperature level was recorded across all cycles. The scatter plot on the right plots Average Final Temperature (x-axis) against Average Final Humidity (y-axis), showing

the inverse relationship between temperature and humidity — as temperature rises, humidity typically falls, increasing fire risk.

Two KPI summary cards are displayed below the charts:

- Average of Final Temp: 29.44°C — the mean temperature across all 300 prediction cycles.
- Sum of Final Hum: 15.35K — the cumulative humidity sum across all cycles, confirming a high overall moisture baseline in this dataset.



Section 4: Data Preparation for Power BI

The Excel file exported by predict_live.py was used directly as the Power BI data source. The following steps were performed in Power BI to prepare the data for visualization:

Import Steps

- Open Power BI Desktop and select Get Data → Excel Workbook.
- Load the forest_fire_predictions.xlsx file and select the predictions sheet.
- In Power Query Editor, verify column data types: Timestamp as Date/Time, Final Temp and Final Hum as Decimal Number, Gas and Flame as Text, Alert and Model as Text, Latitude and Longitude as Decimal Number.
- Apply and close Power Query to load the data into the report canvas.

Calculated Measures Used

```
-- Average Final Temperature
Avg Final Temp = AVERAGE('predictions'[Final Temp (˚C)])
```

```
-- Sum of Humidity
Sum Final Hum = SUM('predictions'[Final Hum (%)])

-- Count of alerts per category
Alert Count = COUNT('predictions'[Alert])
```

Visuals Configuration Summary

Visual Type	Fields Used	Purpose
Pie Chart	Alert (Legend, Values)	Show proportion of each alert category
Donut Chart	Alert (Legend, Values)	Alternative ring view of alert distribution
Bar Chart (Horizontal)	Final Temp, Alert	Show temperature vs. alert type cross-analysis
Gauge	Avg Final Temp	Display average temperature vs. 35°C target
Bar Chart (Horizontal)	Final Temp, Timestamp Count	Show frequency distribution of temperature values
Scatter Plot	Avg Final Temp, Avg Final Hum	Visualize inverse temperature-humidity relationship
KPI Card	Avg Final Temp	Display single-number average temperature
KPI Card	Sum Final Hum	Display total humidity sum across all cycles

Section 5: Key Insights from Dashboard

The Power BI dashboard reveals the following key findings from the 300-cycle prediction dataset:

Alert Distribution Analysis

Nearly 47.33% of all prediction cycles resulted in a NORMAL alert, indicating that a significant portion of readings occurred under safe environmental conditions. However, the remaining 52.67% of cycles recorded some form of fire alert, with CRITICAL: VERIFIED FIRE accounting for 27% alone. This high proportion of critical alerts underscores the volatile nature of the simulated Uttarakhand 2024 fire season conditions embedded in the dataset.

Temperature Analysis

The average Final Temperature of 29.44°C sits below the 35°C danger threshold marked on the gauge. However, the temperature distribution bar chart shows that readings extend well above 40°C, with some cycles reaching 45°C. These elevated readings correspond to the HIGH and FIRE alert categories, confirming that temperature is a primary driver of fire risk classification in this system.

Temperature-Humidity Relationship

The scatter plot confirms the expected inverse relationship between temperature and humidity. The single clustered data point visible in the scatter plot reflects the aggregated average across all 300 cycles (Average Final Temp: 29.44°C, Average Final Hum: approximately 52%). The sum of Final Hum across all cycles is 15.35K, indicating that many individual cycles had high humidity values — consistent with the NORMAL alert majority in the dataset.

Geographic Coverage

All 300 prediction records are geotagged to the Nainital Forest, Uttarakhand region, with latitude values ranging approximately from 29.35 to 29.44 and longitude values from 79.41 to 79.51. This consistent geographic coverage reflects the system's focus on a specific high-risk forest monitoring zone in the Kumaon region of Uttarakhand, India — one of the areas most affected by the 2024 wildfire season.

Section 6: Dashboard Summary Statistics

Metric	Value
Total Prediction Cycles	300
Average Final Temperature	29.44°C
Sum of Final Humidity	15,350% (15.35K)
NORMAL Alert Count	142 (47.33%)
CRITICAL: VERIFIED FIRE Count	81 (27.00%)
FIRE DETECTED (FLAME) Count	74 (24.67%)
EXTREME FIRE RISK (ML) Count	2 (0.67%)
FIRE ALERT (SMOKE + HIGH) Count	1 (0.33%)
Temperature Gauge Target	35°C (elevated fire-risk threshold)
Temperature Gauge Maximum	60°C
Monitored Location	Nainital Forest, Uttarakhand, India
Latitude Range	29.35 – 29.44°N
Longitude Range	79.41 – 79.51°E
Data Source	forest_fire_predictions.xlsx (exported by predict_live.py)
Visualization Tool	Microsoft Power BI Desktop

Outcome

At the end of Milestone 4, the following deliverables were achieved:

- Prediction outputs from the live ML pipeline were stored in a structured Excel and CSV format with 14 columns covering sensor readings, derived features, alert classifications, and geographic coordinates.

- A Microsoft Power BI dashboard was developed with eight visuals across two report pages: alert pie and donut charts, a temperature-alert bar chart, a temperature gauge, a temperature distribution bar chart, a temperature-humidity scatter plot, and two KPI summary cards.
- The dashboard confirmed that 47.33% of prediction cycles resulted in NORMAL conditions while 52.67% triggered some form of fire alert, with CRITICAL: VERIFIED FIRE being the most frequent hazardous category at 27%.
- The average Final Temperature of 29.44°C was visualized against a 35°C danger threshold on the gauge, providing operators with an immediate at-a-glance risk assessment.
- Geographic coordinates embedded in the dataset enable future integration with Power BI Map visuals or Azure Maps for spatial fire-spread visualization.

This milestone completes the analytics and visualization layer of the system. The Power BI dashboard bridges the gap between raw sensor data and actionable intelligence, enabling forest management authorities and disaster response teams to monitor environmental conditions, track fire alert trends, and make data-driven decisions for early intervention.